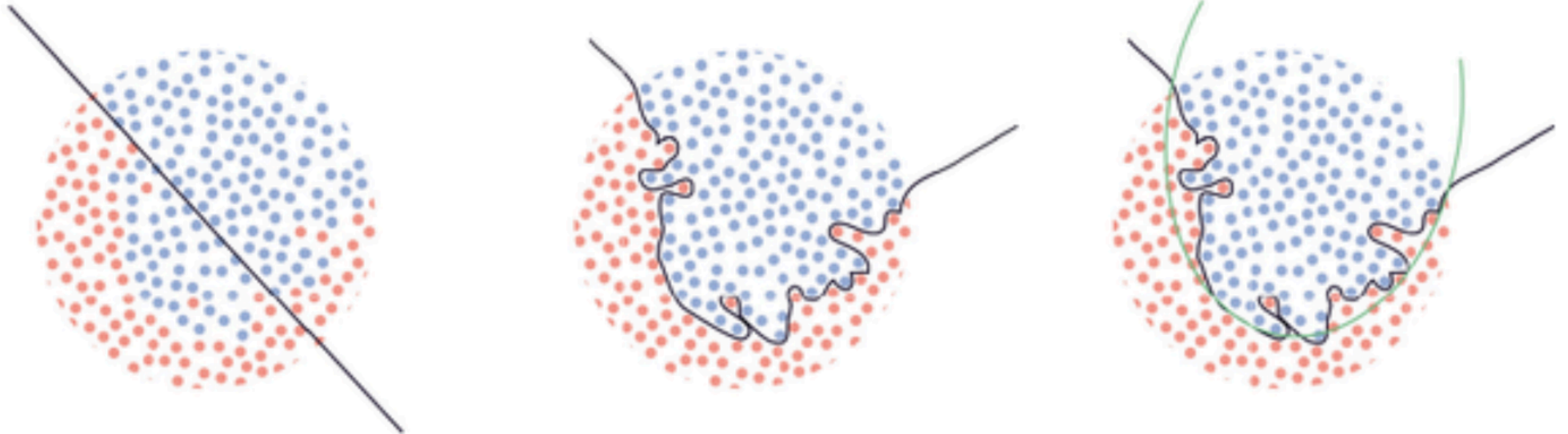


Regularization

CSC 487: Deep Learning
Instructor: Jonathan Ventura

Regularization reduces overfitting



L2 regularization (weight decay)

- Can use **weight regularization** to control overfitting.
- Let w_{ijk} be the weight for layer i at matrix location j, k
- L2 Regularization:

$$L_{total} = L + \lambda \sum_i \sum_j \sum_k w_{ijk}^2$$

where λ controls the strength of the regularization (could be 0.01 for example)

- Effect on loss gradient:

$$\frac{\partial L_{total}}{\partial w_{ijk}} = \frac{\partial L}{\partial w_{ijk}} + \lambda w_{ijk} \quad \text{(subtract a small multiple of the weight at each update step)}$$

L1 regularization

- L1 Regularization:

$$L_{total} = L + \lambda \sum_i \sum_j \sum_k |w_{ijk}|$$

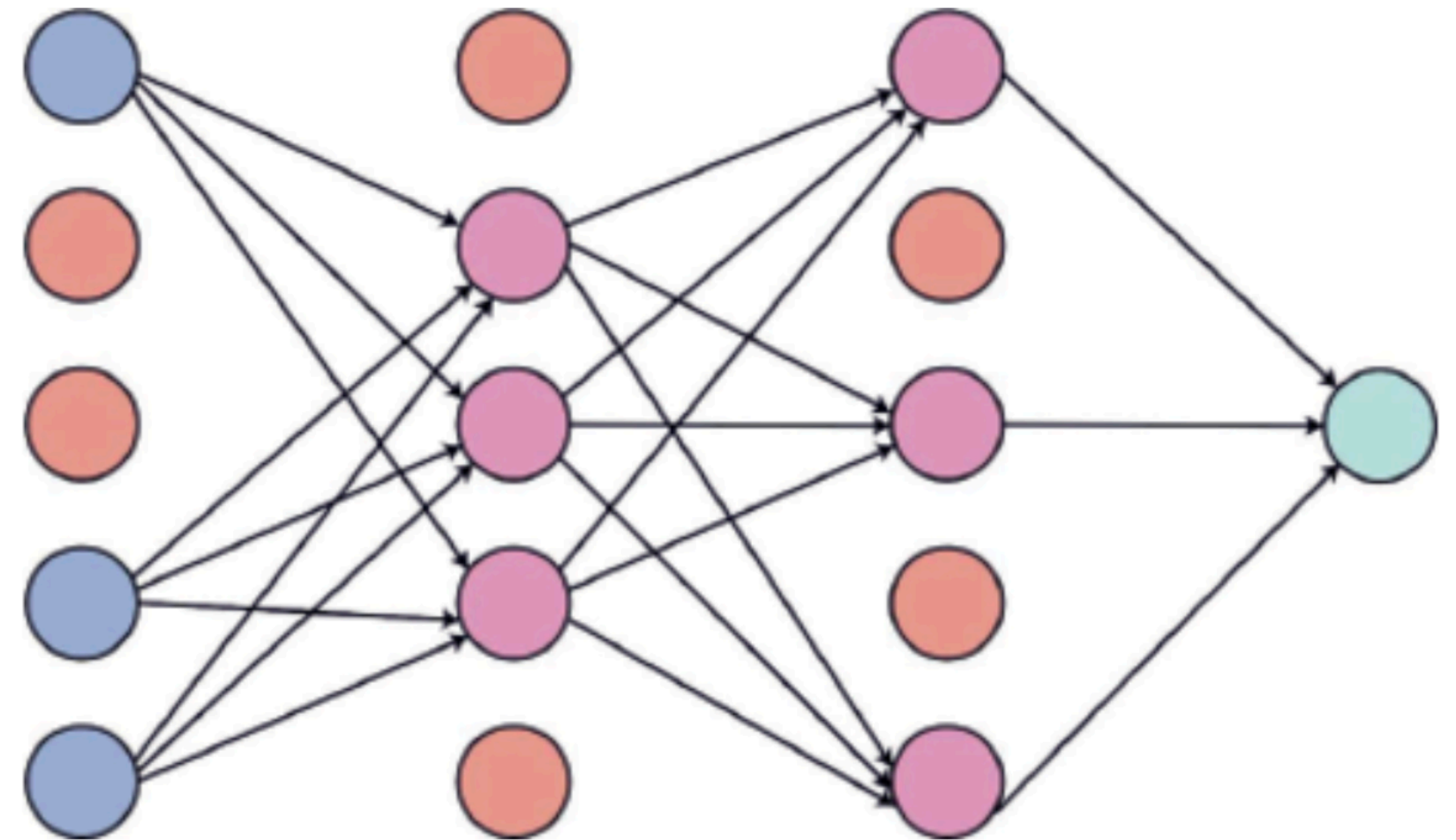
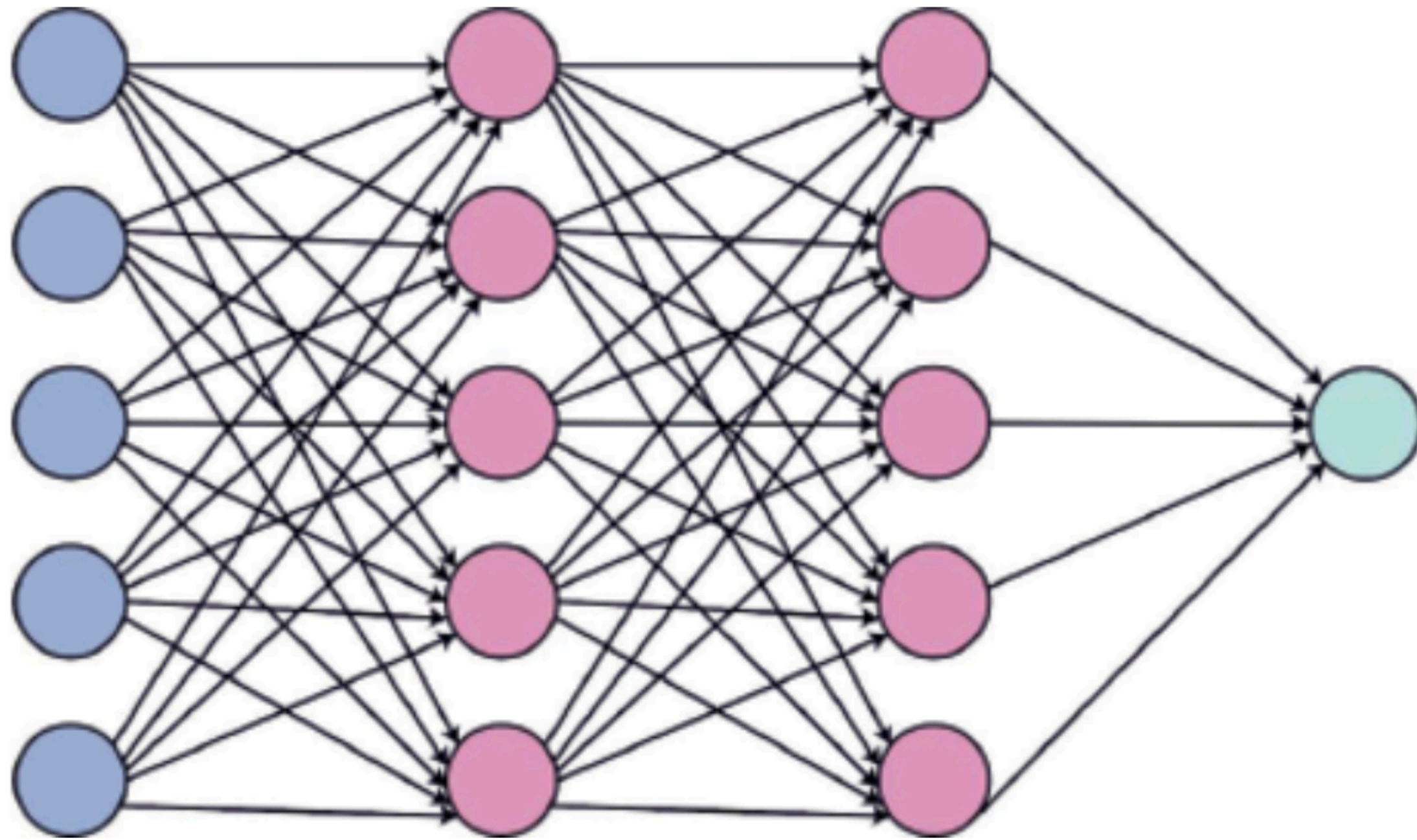
- Effect on loss gradient:

$$\frac{\partial L_{total}}{\partial w_{ijk}} = \frac{\partial L}{\partial w_{ijk}} + \lambda \cdot \text{sign}(w_{ijk})$$

(reduce the weight by a small constant at each step)

Dropout regularization

- Randomly drop neurons during training with probability p



- Scale neurons by $1/(1 - p)$ when activated during training.
- Activate all neurons during inference

Case study: California Housing Dataset

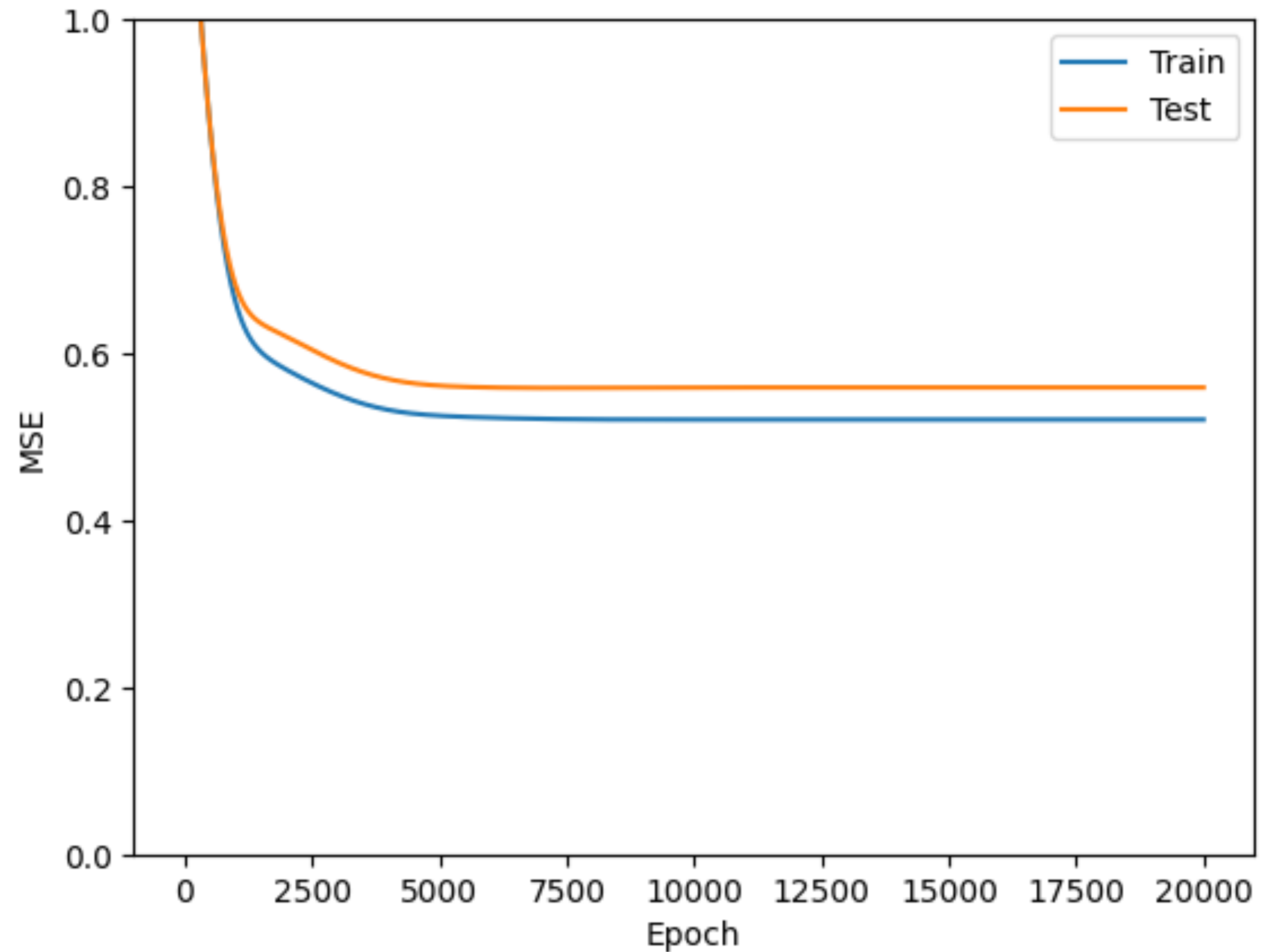
- Data from 1990 U.S. census, per census block group.

Med. House Price	Med. Income	House Age	Avg. Rooms	Avg. Bedrooms	Population	Avg. Occupancy
4.526	8.32	41	6.98	1.02	322	2.56
3.585	8.30	21	62.4	0.972	240	2.11
3.521	7.26	52	8.29	1.07	496	2.80
...	...	...	...	...	...	...

- Created random 90/10 train/test split.
- Use mean squared error as loss function.

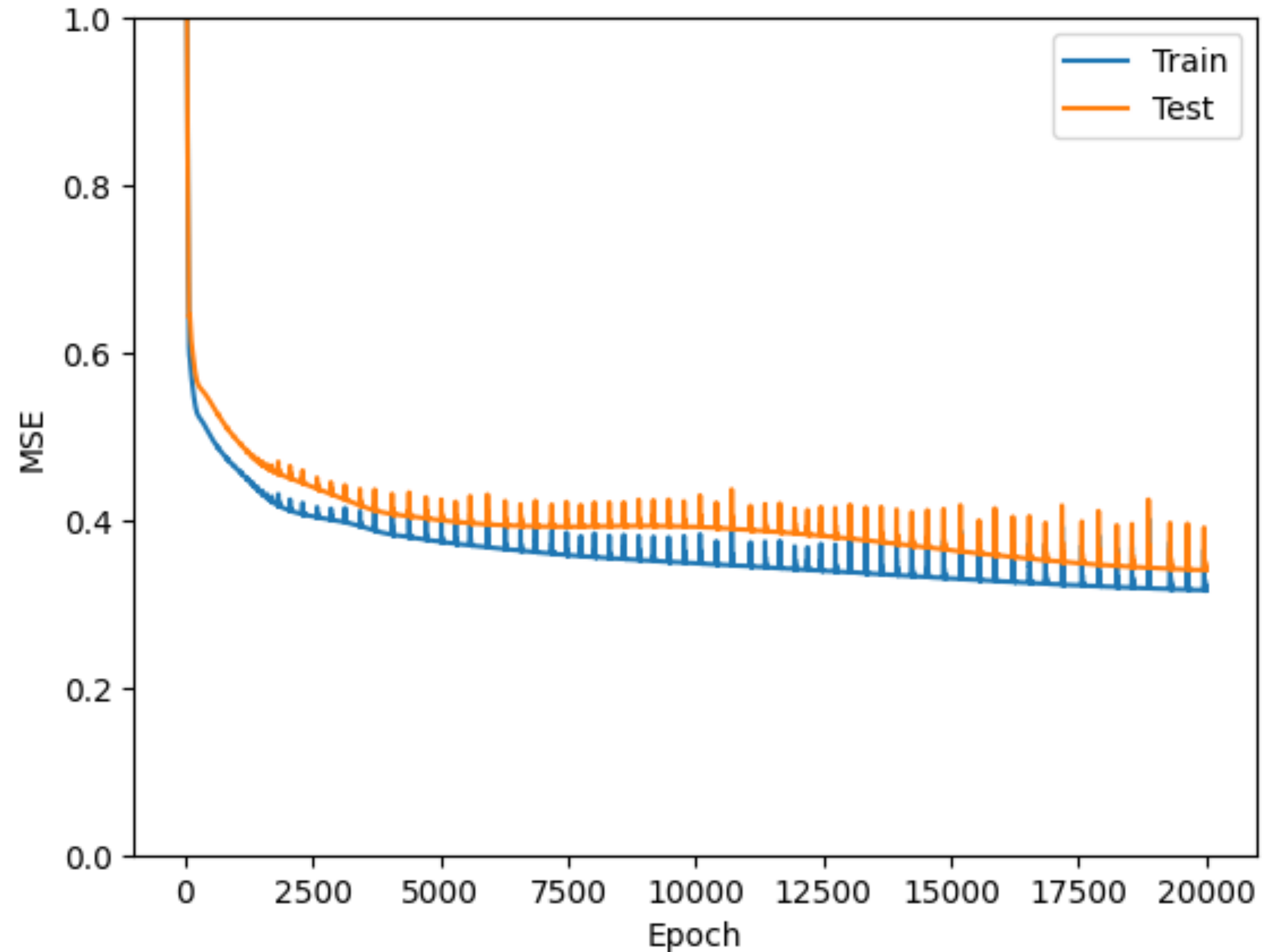
Linear model

- 9 parameters
- Best train MSE: 0.521
- Best test MSE: 0.558



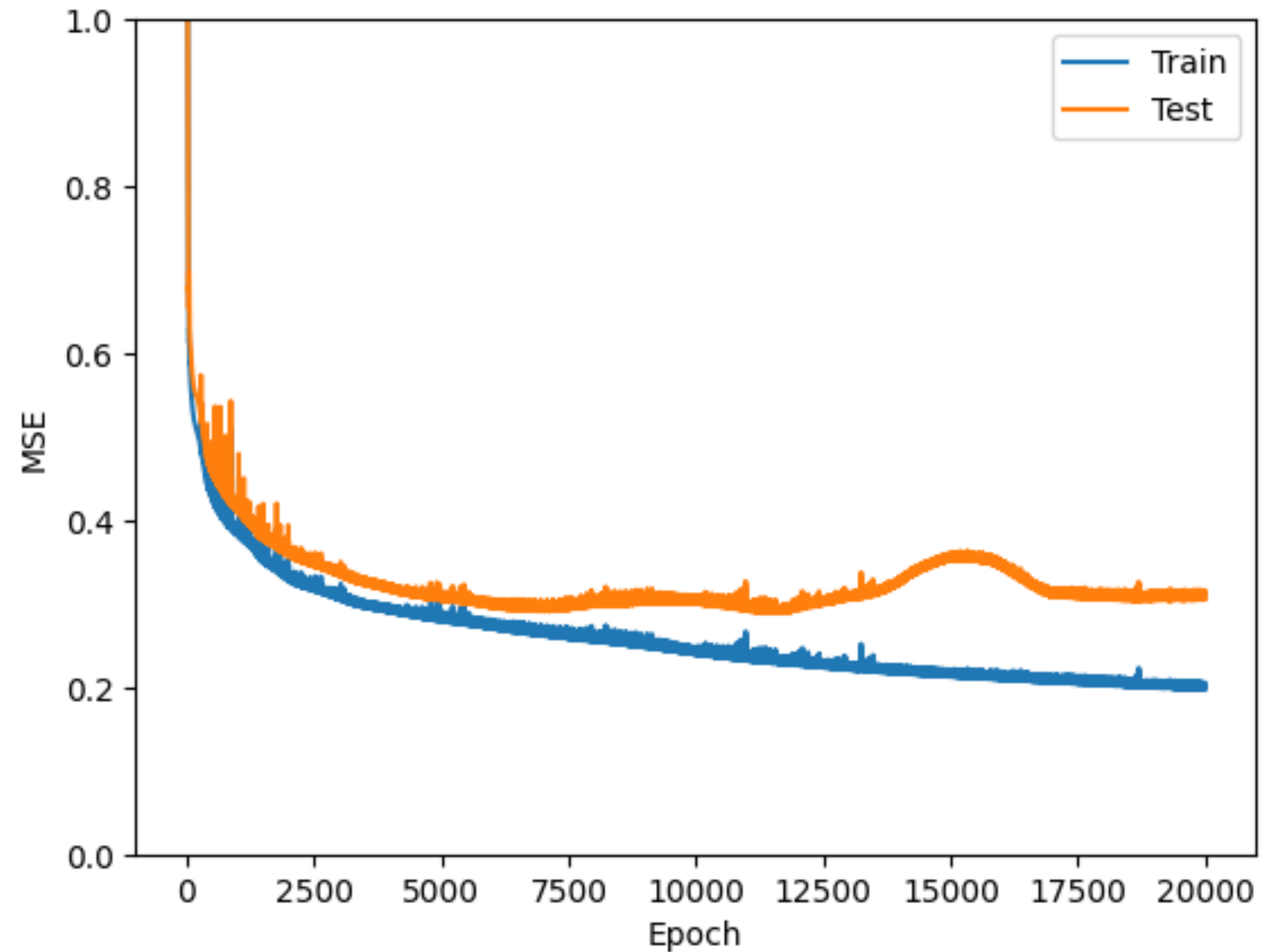
1 hidden layer, width 100

- # parameters: 1,001
- Best train MSE: 0.316
- Best test MSE: 0.340



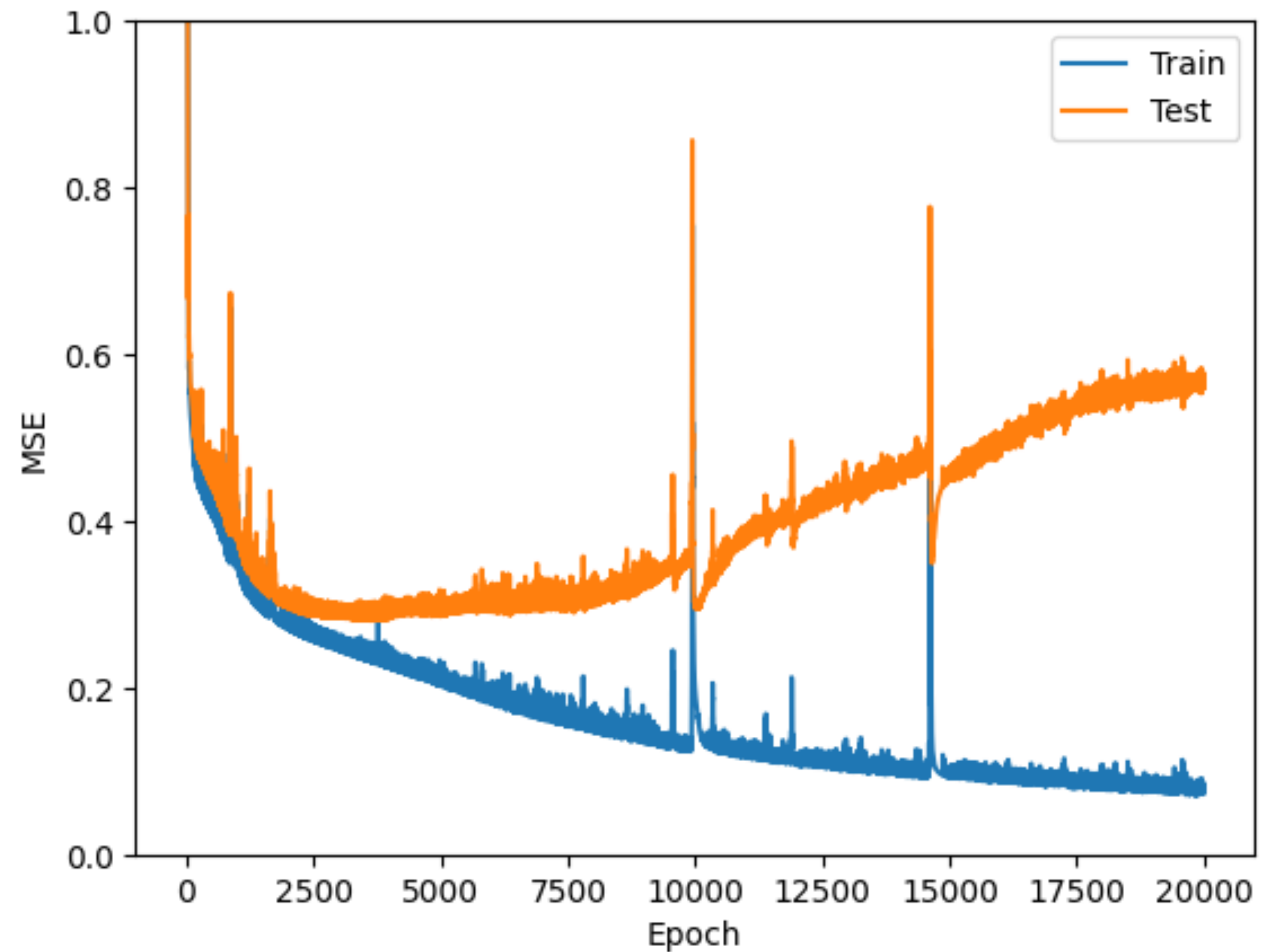
2 hidden layers, width 100

- # parameters: 11,101
- Best train MSE: 0.228
- Best test MSE: 0.289



3 hidden layers, width 100

- # parameters: 21,201
- Best train MSE: 0.231
- Best test MSE: 0.280



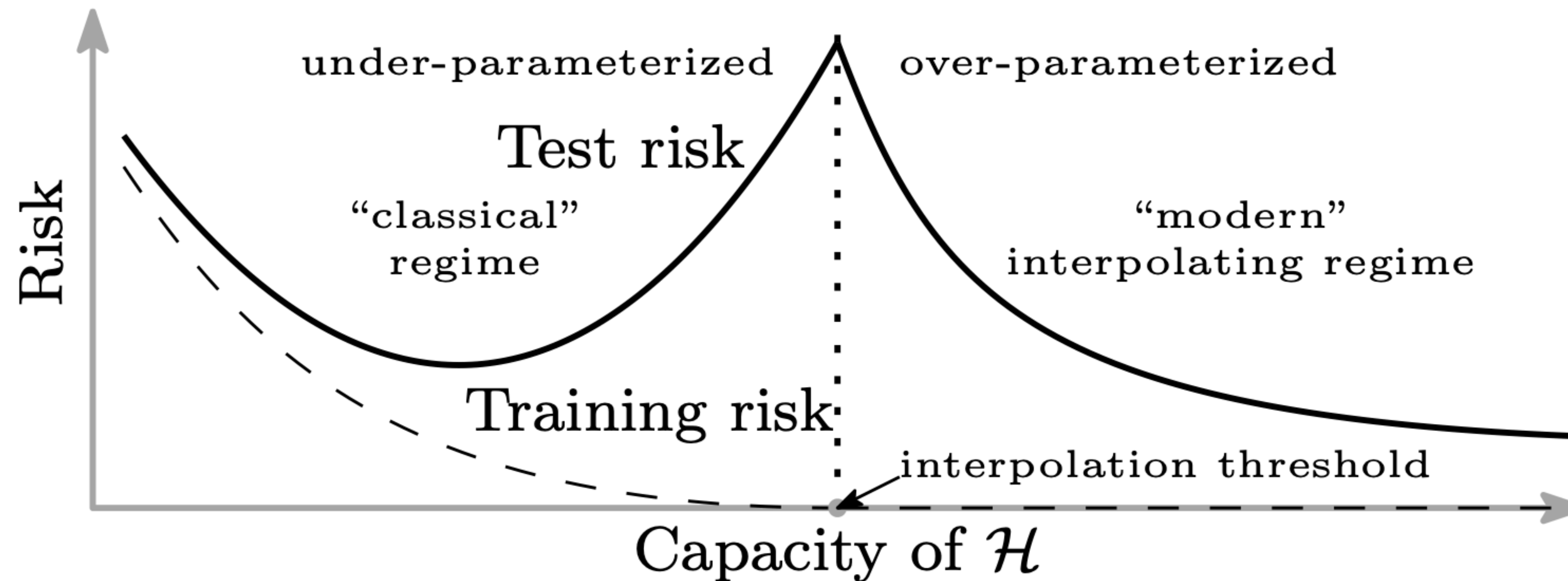
3 hidden layers, width 100 with L2 regularization

L2	Train loss	Test loss
0	0.231	0.280
0.00001	0.205	0.264
0.0001	0.230	0.260
0.001	0.429	0.468
0.01	0.503	0.542

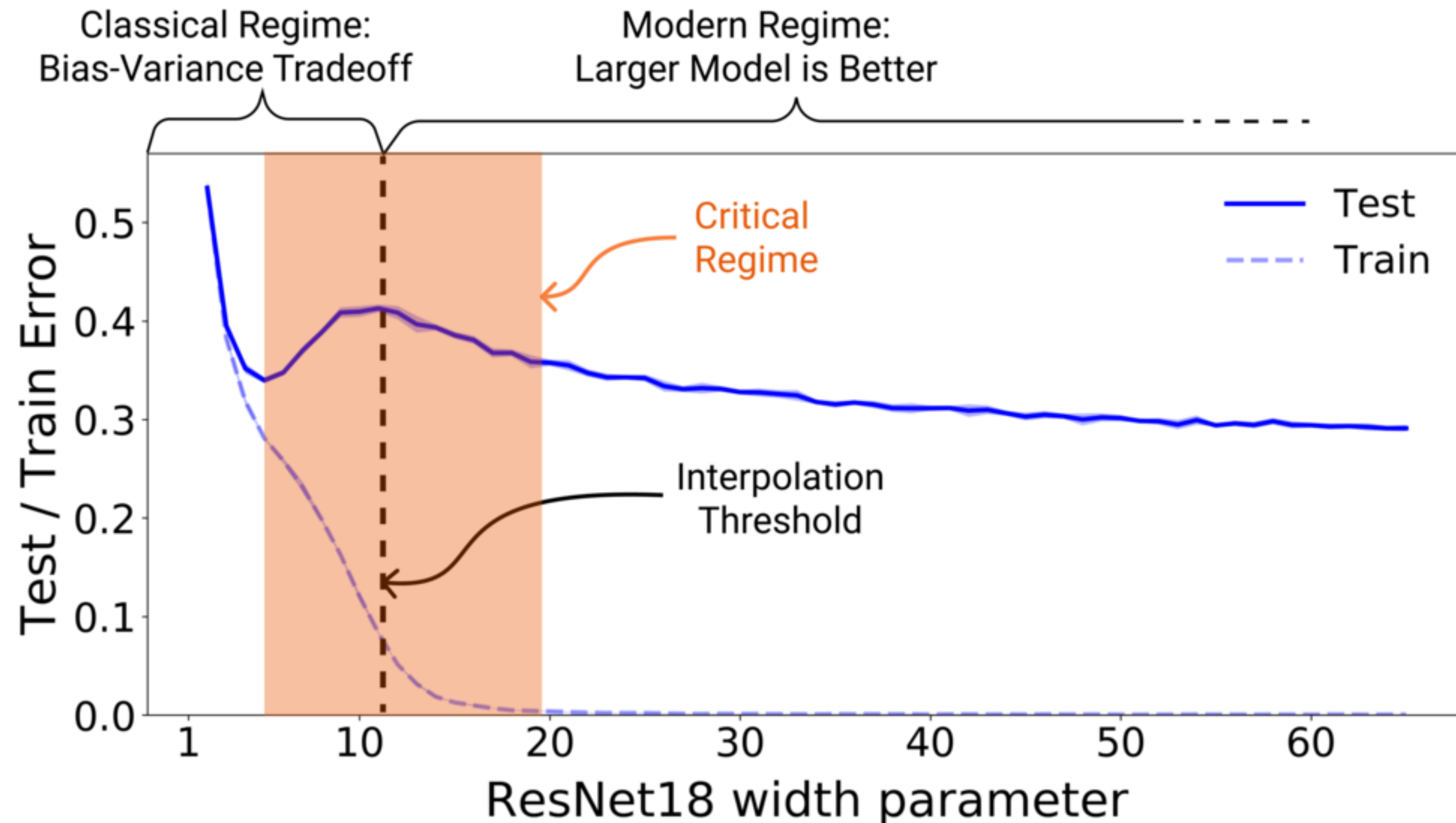
Width 100, 0.0001 L2 regularization

# Hidden Layers	Train loss	Test loss
3	0.231	0.280
4	0.212	0.245
5	0.202	0.234
6	0.176	0.226
7	0.189	0.218

“Double descent”: moving past the bias-variance tradeoff



Deep Double Descent: Where bigger models and more data hurt



Rethinking Parameter Counting in Deep Models: Effective Dimensionality Revisited

